

Research Profile and Selected Project

Yohannes Alealign Biresaw

BSc in Electrical and Computer Engineering | GPA 3.63/4.0

Kuwait | yohannes.sch.ca@gmail.com | github.com/john121319

Research Direction

My long-term research interest is secure and trustworthy intelligent systems. I am interested in artificial intelligence and machine learning, cybersecurity, applied cryptography, privacy-preserving computation, secure distributed systems, and medical-image analysis. The common thread is the development of systems that remain accurate, robust, secure, and privacy-aware when they operate on sensitive, distributed, or heterogeneous data.

My completed federated-learning project is the strongest current evidence of my research preparation, but it does not limit my future direction to one algorithm or one application area. The research skills developed through the project are transferable to a wider range of AI, machine-learning, cybersecurity, and privacy problems.

Completed Research Project

Evaluating FedAvg and FedProx for Federated Classification of Pneumonia-Associated Lung Opacity Under Non-IID Data

Using the RSNA Pneumonia Detection Challenge dataset, I built a patient-aware pipeline and compared centralized learning, FedAvg, and FedProx under IID and controlled non-IID conditions.

- 26,684 chest-radiograph examinations linked to 11,452 original patients, with zero patient overlap across training, validation, and test splits.
- Five simulated clients under IID, moderate non-IID, and severe non-IID conditions.
- Three random seeds, validation-only checkpoint and threshold selection, and one frozen final test evaluation of 18 selected models.

Main Results

Condition	ROC-AUC	PR-AUC	Balanced accuracy	F1-score
Centralized	0.8295	0.5800	0.7510	0.5745
FedAvg IID	0.8179	0.5605	0.7453	0.5695
FedAvg moderate	0.8131	0.5553	0.7355	0.5541
FedAvg severe	0.8070	0.5440	0.7312	0.5529
FedProx moderate	0.8112	0.5505	0.7375	0.5608
FedProx severe	0.8076	0.5428	0.7335	0.5581

Increasing client heterogeneity reduced federated performance, particularly PR-AUC. FedAvg retained a small PR-AUC advantage, while FedProx achieved slightly higher balanced accuracy and F1-score and reduced client and global update magnitudes by approximately one-half.

Central interpretation: optimization stability and predictive generalization were related but not identical. FedProx clearly limited model movement, yet neither method consistently dominated across all predictive metrics.

Research Preparation Demonstrated

- Patient-aware medical-image preparation, leakage prevention, and DICOM preprocessing.
- CNN development and evaluation using TensorFlow/Keras.
- Federated aggregation, controlled IID/non-IID partitioning, and client-drift analysis.
- Multi-seed experimentation, validation-based model selection, and a frozen final-test protocol.
- Experiment tracking, reproducible reporting, and interpretation of mixed findings without overstating the evidence.

Broader Graduate Research Interests

- Artificial intelligence and machine learning: robust learning, generalization, calibration, trustworthy AI, and real-world applications.
- Cybersecurity and AI security: detection and defense, adversarial robustness, model integrity, and secure system design.
- Cryptography and privacy: applied cryptography, privacy-preserving computation, secure aggregation, and protection of distributed learning systems.
- Secure distributed systems: privacy, integrity, communication, and threat-aware collaboration across devices or institutions.
- Medical and high-stakes AI: uncertainty, fairness, careful evaluation, and responsible use.

Possible MSc Directions

- Trustworthy and privacy-preserving machine learning under heterogeneous or distributed data.
- AI for cybersecurity, including anomaly detection, threat intelligence, and robust classification.
- Security of machine-learning pipelines, including adversarial behavior, poisoning, privacy leakage, and model integrity.
- Applied cryptographic mechanisms for collaborative or distributed learning.
- Reliable medical-image analysis, calibration, uncertainty, and external validation.

I would welcome a thesis topic shaped by the advisor's research program. My priority is to work on one focused question that fits the group while building deeper foundations in mathematics, research methods, security, and machine learning.

Graduate Goal

I am seeking a thesis-based master's opportunity where I can contribute to carefully designed research and grow toward work at the intersection of intelligent systems, security, privacy, and dependable real-world deployment.